

INDUCTION AND INDUCTANCE

CHAPTER

30



30-1

WHAT IS PHYSICS?

In Chapter 29 we discussed the fact that a current produces a magnetic field. That fact came as a surprise to the scientists who discovered the effect. Perhaps even more surprising was the discovery of the reverse effect: A magnetic field can produce an electric field that can drive a current. This link between a magnetic field and the electric field it produces (*induces*) is now called *Faraday's law of induction*.



30-2 Two Experiments

Let us examine two simple experiments to prepare for our discussion of Faraday's law of induction.

First Experiment. Figure 30-1 shows a conducting loop connected to a sensitive ammeter. Because there is no battery or other source of emf included, there is no current in the circuit. However, if we move a bar magnet toward the loop, a current suddenly appears in the circuit. The current disappears when the magnet stops. If we then move the magnet away, a current again suddenly appears, but now in the opposite direction. If we experimented for a while, we would discover the following:

1. A current appears only if there is relative motion between the loop and the magnet (one must move relative to the other); the current disappears when the relative motion between them ceases.

The magnet's motion creates a current in the loop.

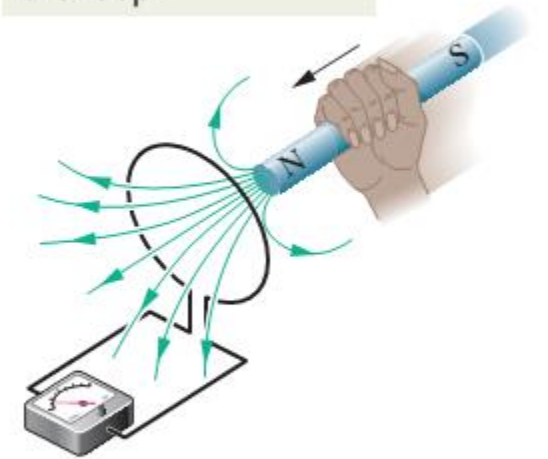
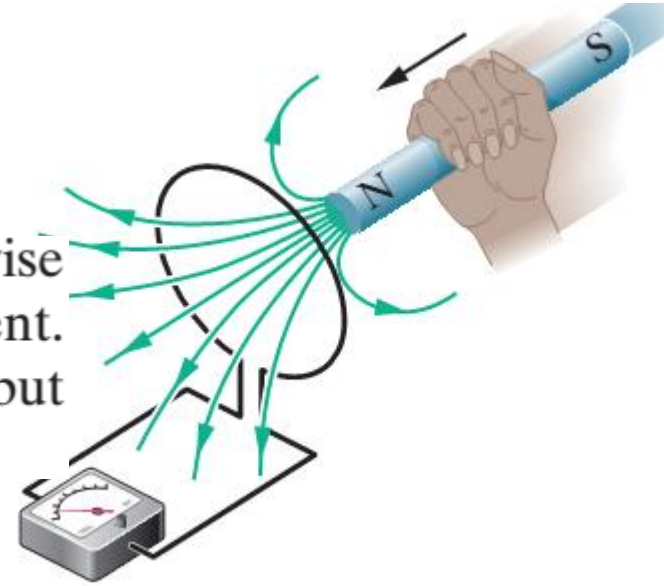


Fig. 30-1 An ammeter registers a current in the wire loop when the magnet is moving with respect to the loop.

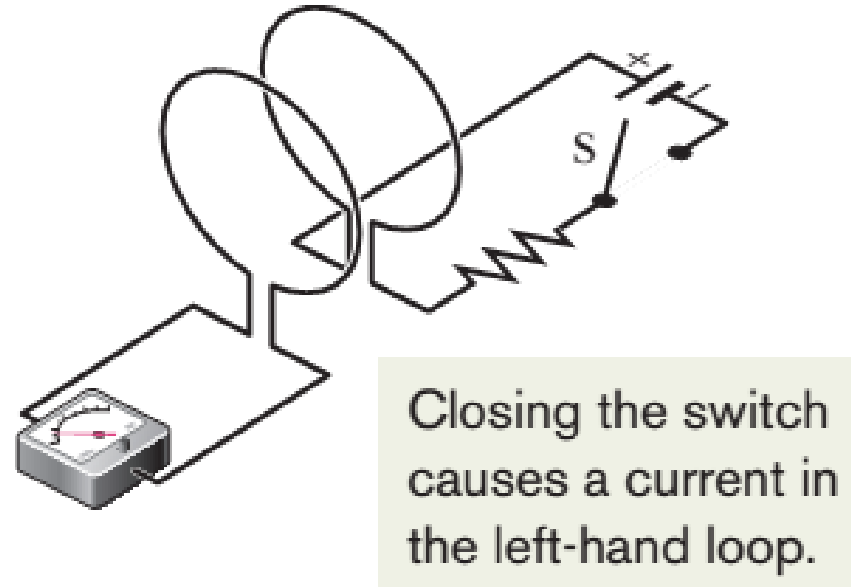


1. A current appears only if there is relative motion between the loop and the magnet (one must move relative to the other); the current disappears when the relative motion between them ceases.
2. Faster motion produces a greater current.
3. If moving the magnet's north pole toward the loop causes, say, clockwise current, then moving the north pole away causes counterclockwise current. Moving the south pole toward or away from the loop also causes currents, but in the reversed directions.

The current produced in the loop is called an **induced current**; the work done per unit charge to produce that current (to move the conduction electrons that constitute the current) is called an **induced emf**; and the process of producing the current and emf is called **induction**.



Second Experiment.



with the two conducting loops close to each other but not touching. If we close switch S, to turn on a current in the right-hand loop, the meter suddenly and briefly registers a current—an induced current—in the left-hand loop. If we then open the switch, another sudden and brief induced current appears in the left-hand loop, but in the opposite direction. We get an induced current (and thus an induced emf) only when the current in the right-hand loop is changing (either turning on or turning off) and not when it is constant (even if it is large).

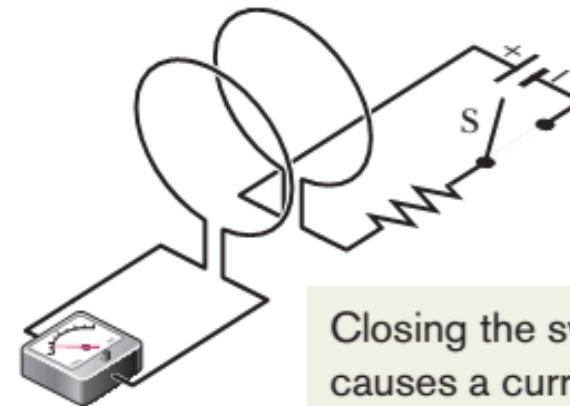
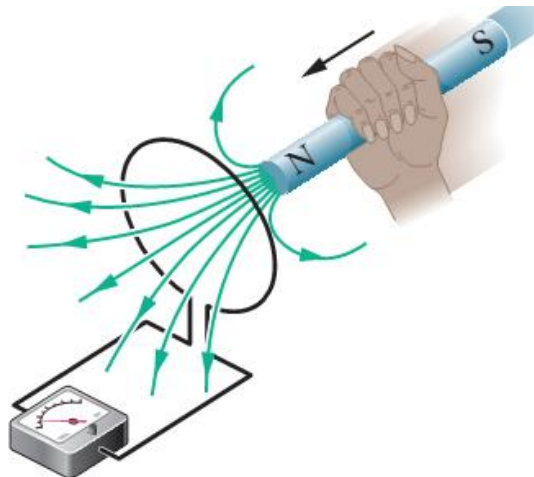


30-3 Faraday's Law of Induction

Faraday realized that an emf and a current can be induced in a loop, as in our two experiments, by changing the *amount of magnetic field* passing through the loop. He further realized that the “amount of magnetic field” can be visualized in terms of the magnetic field lines passing through the loop. **Faraday's law of induction**, stated in terms of our experiments, is this:



An emf is induced in the loop at the left in Figs. 30-1 and 30-2 when the number of magnetic field lines that pass through the loop is changing.



Closing the switch causes a current in the left-hand loop.



magnetic flux

$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad (\text{magnetic flux through area } A).$$

$$\Phi_B = BA \quad (\vec{B} \perp \text{area } A, \vec{B} \text{ uniform}). \quad (30-2)$$

$$1 \text{ weber} = 1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2.$$

With the notion of magnetic flux, we can state Faraday's law in a more quantitative and useful way:



The magnitude of the emf $\mathcal{E}$ induced in a conducting loop is equal to the rate at which the magnetic flux Φ_B through that loop changes with time.





The magnitude of the emf $\mathcal{E}$ induced in a conducting loop is equal to the rate at which the magnetic flux Φ_B through that loop changes with time.

$$\mathcal{E} = - \frac{d\Phi_B}{dt} \quad (\text{Faraday's law}),$$

with the minus sign indicating that opposition. We often neglect the minus sign seeking only the magnitude of the induced emf.

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad (\text{coil of } N \text{ turns}). \quad (30-5)$$



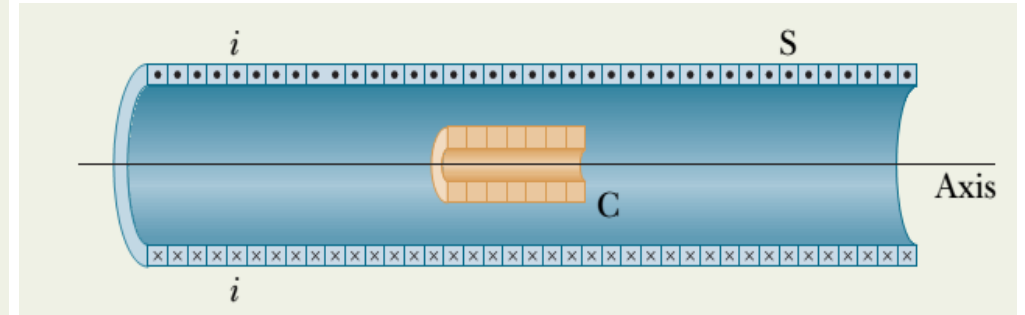
Here are the general means by which we can change the magnetic flux through a coil:

1. Change the magnitude B of the magnetic field within the coil.
2. Change either the total area of the coil or the portion of that area that lies within the magnetic field (for example, by expanding the coil or sliding it into or out of the field).
3. Change the angle between the direction of the magnetic field $\vec{B}$ and the plane of the coil (for example, by rotating the coil so that field $\vec{B}$ is first perpendicular to the plane of the coil and then is along that plane).



Sample Problem

The long solenoid S shown (in cross section) in Fig. 30-3 has 220 turns/cm and carries a current $i = 1.5$ A; its diameter D is 3.2 cm. At its center we place a 130-turn closely packed coil C of diameter $d = 2.1$ cm. The current in the solenoid is reduced to zero at a steady rate in 25 ms. What is the magnitude of the emf that is induced in coil C while the current in the solenoid is changing?



Now we can write

$$\Phi_{B,i} = BA = (\mu_0 in)A$$

$$A \text{ is } \frac{1}{4}\pi d^2$$

$$\begin{aligned} &= (4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(1.5 \text{ A})(22\,000 \text{ turns/m}) \\ &\quad \times (3.464 \times 10^{-4} \text{ m}^2) \\ &= 1.44 \times 10^{-5} \text{ Wb.} \end{aligned}$$

$$\begin{aligned} \frac{d\Phi_B}{dt} &= \frac{\Delta\Phi_B}{\Delta t} = \frac{\Phi_{B,f} - \Phi_{B,i}}{\Delta t} \\ &= \frac{(0 - 1.44 \times 10^{-5} \text{ Wb})}{25 \times 10^{-3} \text{ s}} \\ &= -5.76 \times 10^{-4} \text{ Wb/s} = -5.76 \times 10^{-4} \text{ V.} \end{aligned}$$

$$\begin{aligned} \mathcal{E} &= N \frac{d\Phi_B}{dt} = (130 \text{ turns})(5.76 \times 10^{-4} \text{ V}) \\ &= 7.5 \times 10^{-2} \text{ V} = 75 \text{ mV.} \end{aligned}$$

(Answer)



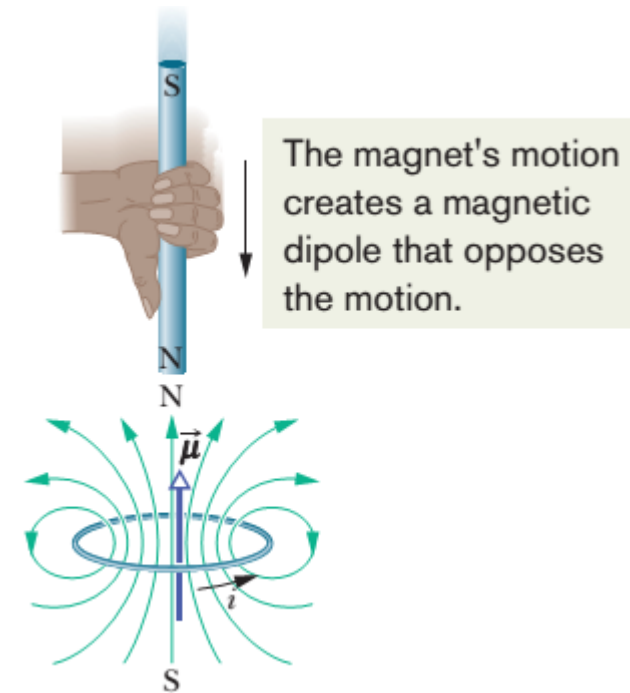
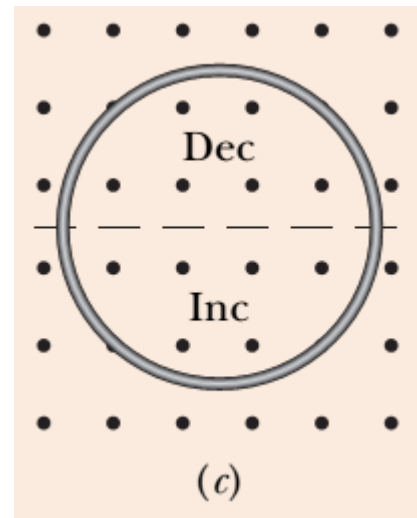
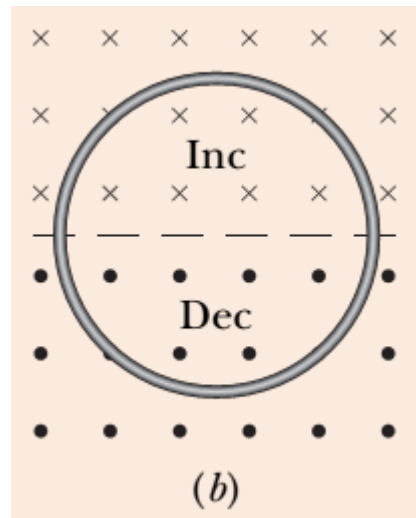
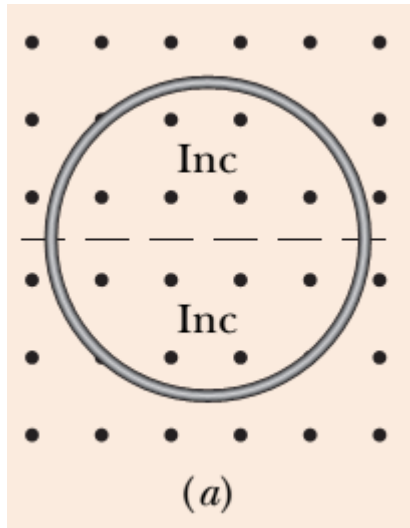
30-4 Lenz's Law



An induced current has a direction such that the magnetic field due to *the current* opposes the change in the magnetic flux that induces the current.

1. *Opposition to Pole Movement.*

2. *Opposition to Flux Change.*

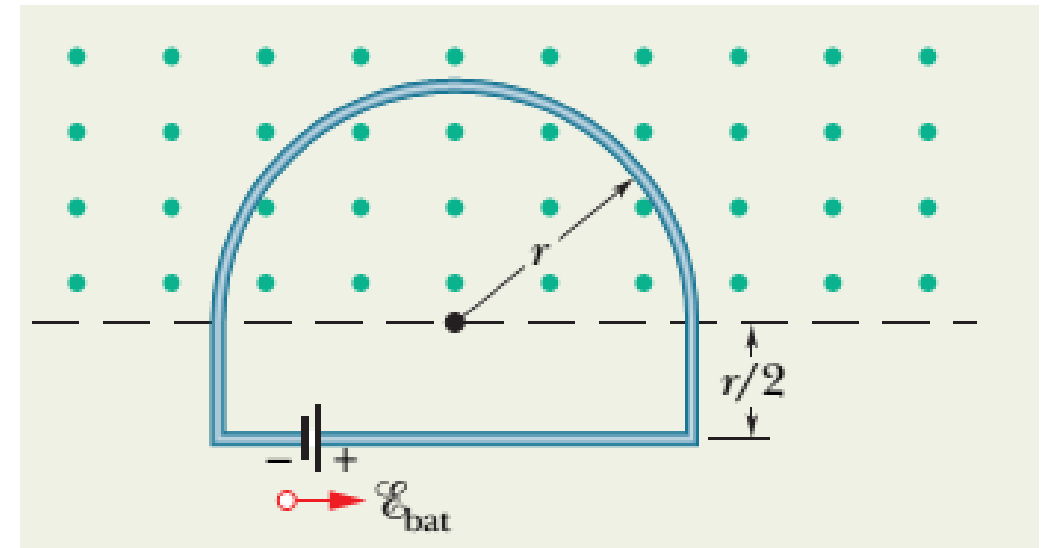


Sample Problem

Figure 30-6 shows a conducting loop consisting of a half-circle of radius $r = 0.20$ m and three straight sections. The half-circle lies in a uniform magnetic field $\vec{B}$ that is directed out of the page; the field magnitude is given by $B = 4.0t^2 + 2.0t + 3.0$, with B in teslas and t in seconds. An ideal battery with emf $\mathcal{E}_{\text{bat}} = 2.0$ V is connected to the loop. The resistance of the loop is $2.0\ \Omega$.

(a) What are the magnitude and direction of the emf $\mathcal{E}_{\text{ind}}$ induced around the loop by field $\vec{B}$ at $t = 10$ s?

(b) What is the current in the loop at $t = 10$ s?



$$\mathcal{E}_{\text{ind}} = \frac{d\Phi_B}{dt} = \frac{d(BA)}{dt} = A \frac{dB}{dt}.$$

A in this equation is $\frac{1}{2}\pi r^2$.

$$\mathcal{E}_{\text{ind}} = A \frac{dB}{dt} = \frac{\pi r^2}{2} \frac{d}{dt} (4.0t^2 + 2.0t + 3.0)$$

$$= \frac{\pi r^2}{2} (8.0t + 2.0).$$



At $t = 10$ s, then,

$$\begin{aligned}\mathcal{E}_{\text{ind}} &= \frac{\pi (0.20 \text{ m})^2}{2} [8.0(10) + 2.0] \\ &= 5.152 \text{ V} \approx 5.2 \text{ V}.\end{aligned}$$

$$\begin{aligned}i &= \frac{\mathcal{E}_{\text{net}}}{R} = \frac{\mathcal{E}_{\text{ind}} - \mathcal{E}_{\text{bat}}}{R} \\ &= \frac{5.152 \text{ V} - 2.0 \text{ V}}{2.0 \Omega} = 1.58 \text{ A} \approx 1.6 \text{ A}. \quad (\text{Answer})\end{aligned}$$

Sample Problem

797

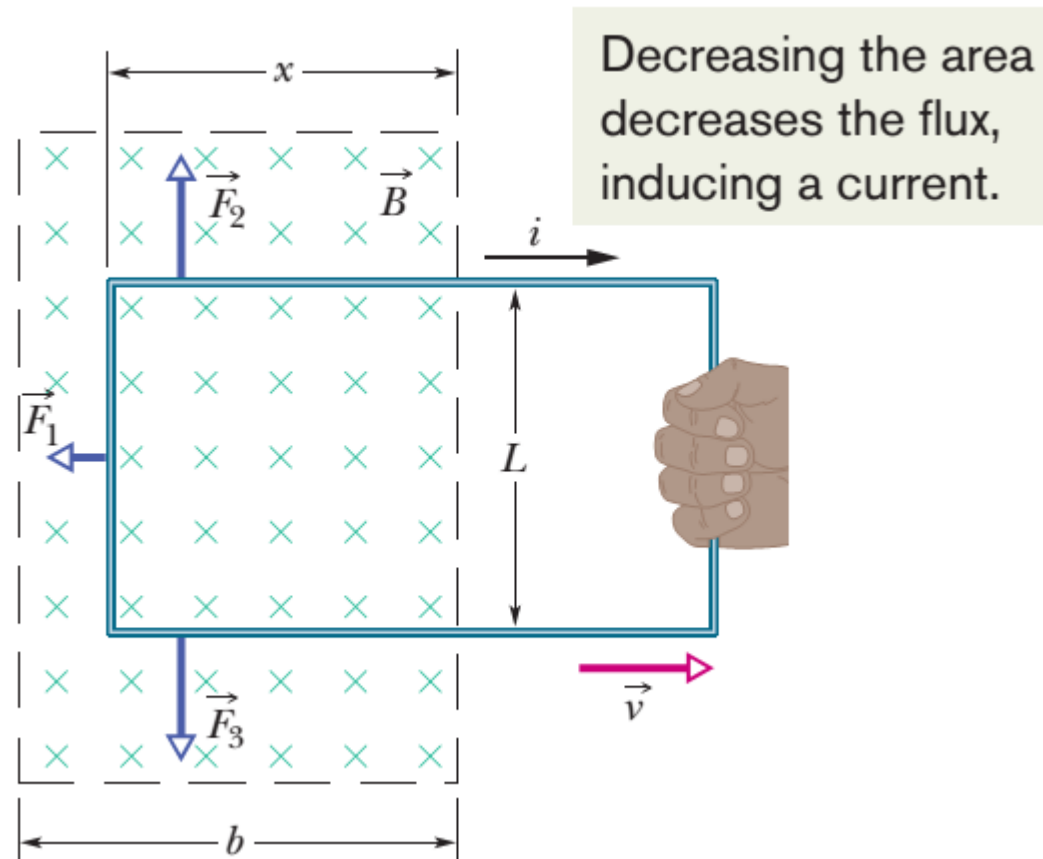
Induced emf due to a changing nonuniform B field

Figure 30-7 shows a rectangular loop of wire immersed in a nonuniform and varying magnetic field $\vec{B}$ that is perpendicular to and directed into the page. The field's magnitude is given by $B = 4t^2x^2$, with B in teslas, t in seconds,

and x in meters. (Note that the function depends on *both* time and position.) The loop has width $W = 3.0$ m and height $H = 2.0$ m. What are the magnitude and direction of the induced emf $\mathcal{E}$ around the loop at $t = 0.10$ s?



30-5 Induction and Energy Transfers



As you will see, to pull the loop at a constant velocity $\vec{v}$, you must apply a constant force $\vec{F}$ to the loop because a magnetic force of equal magnitude but opposite direction acts on the loop to oppose you. From Eq. 7-48, the rate at which you do work—that is, the power—is then

$$P = Fv, \quad (30-6)$$



$$\Phi_B = BA = BLx. \quad (30-7)$$

$$\mathcal{E} = \frac{d\Phi_B}{dt} = \frac{d}{dt} BLx = BL \frac{dx}{dt} = BLv, \quad (30-8)$$

replaced dx/dt with v

$$i = \mathcal{E}/R$$

$$i = \frac{BLv}{R}. \quad (30-9)$$

$$\vec{F}_d = i\vec{L} \times \vec{B}. \quad (30-10)$$

$$F = \frac{B^2 L^2 v}{R}. \quad (30-12)$$



$$P = Fv = \frac{B^2 L^2 v^2}{R} \quad (\text{rate of doing work}). \quad (30-13)$$

$$P = i^2 R. \quad (30-14)$$

$$P = \left(\frac{BLv}{R} \right)^2 R = \frac{B^2 L^2 v^2}{R} \quad (\text{thermal energy rate}), \quad (30-15)$$

which is exactly equal to the rate at which you are doing work





30-6 Induced Electric Fields



Fig. 30-11 (a) If the magnetic field increases at a steady rate, a constant induced current appears, as shown, in the copper ring of radius r . (b) An induced electric field exists even when the ring is removed; the electric field is shown at four points.



A changing magnetic field produces an electric field.



A Reformulation of Faraday's Law



A changing magnetic field produces an electric field.

Consider a particle of charge q_0 moving around the circular path of Fig. 30-11*b*. The work W done on it in one revolution by the induced electric field is $W = \mathcal{E}q_0$, where $\mathcal{E}$ is the induced emf—that is, the work done per unit charge in moving the test charge around the path. From another point of view, the work is

$$W = \int \vec{F} \cdot d\vec{s} = (q_0 E)(2\pi r), \quad (30-16)$$

$$\mathcal{E} = 2\pi r E. \quad (30-17)$$

Next we rewrite Eq. 30-16 to give a more general expression for the work done on a particle of charge q_0 moving along any closed path:

$$W = \oint \vec{F} \cdot d\vec{s} = q_0 \oint \vec{E} \cdot d\vec{s}. \quad (30-18)$$

Substituting $\mathcal{E}q_0$ for W , we find that

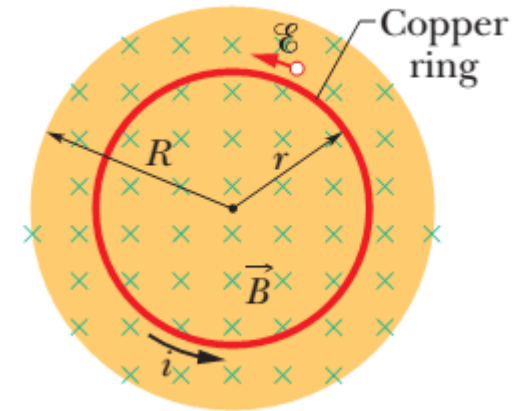
$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s}. \quad (30-19)$$



$$\mathcal{E} = -d\Phi_B/dt$$

Faraday's law

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\Phi_B}{dt} \quad (\text{Faraday's law}). \quad (30-20)$$



A New Look at Electric Potential



Electric potential has meaning only for electric fields that are produced by static charges; it has no meaning for electric fields that are produced by induction.

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}. \quad (30-21)$$

V_i and V_f are identical, and Eq. 30-21 reduces to

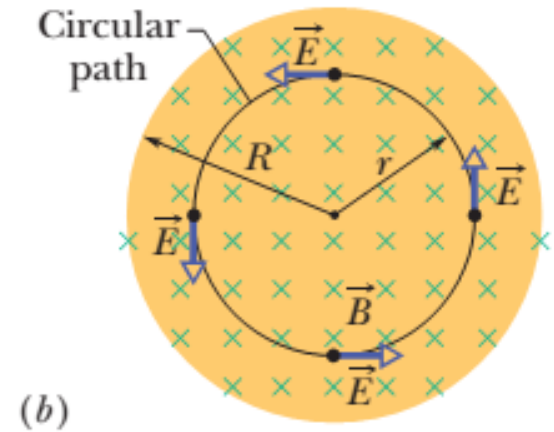
$$\oint \vec{E} \cdot d\vec{s} = 0. \quad (30-22)$$



In Fig. 30-11*b*, take $R = 8.5$ cm and $dB/dt = 0.13$ T/s.

(a) Find an expression for the magnitude E of the induced electric field at points within the magnetic field, at radius r from the center of the magnetic field. Evaluate the expression for $r = 5.2$ cm.

(b) Find an expression for the magnitude E of the induced electric field at points that are outside the magnetic field, at radius r from the center of the magnetic field. Evaluate the expression for $r = 12.5$ cm.



$$\oint \vec{E} \cdot d\vec{s} = \oint E ds = E \oint ds = E(2\pi r). \quad (30-23)$$

$$\Phi_B = BA = B(\pi r^2). \quad (30-24)$$

$$E(2\pi r) = (\pi r^2) \frac{dB}{dt}$$

$$E = \frac{r}{2} \frac{dB}{dt}. \quad (\text{Answer}) \quad (30-25)$$



Equation 30-25 gives the magnitude of the electric field at any point for which $r \leq R$ (that is, within the magnetic field). Substituting given values yields, for the magnitude of $\vec{E}$ at $r = 5.2 \text{ cm}$,

$$E = \frac{(5.2 \times 10^{-2} \text{ m})}{2} (0.13 \text{ T/s})$$

$$= 0.0034 \text{ V/m} = 3.4 \text{ mV/m}.$$

Calculations: We can now write

$$\Phi_B = BA = B(\pi R^2). \quad (30-26)$$

Substituting this and Eq. 30-23 into Eq. 30-20 (without the minus sign) and solving for E yield

$$E = \frac{R^2}{2r} \frac{dB}{dt}. \quad (\text{Answer}) \quad (30-27)$$



$\vec{E}$ at $r = 12.5$ cm:

$$\begin{aligned} E &= \frac{(8.5 \times 10^{-2} \text{ m})^2}{(2)(12.5 \times 10^{-2} \text{ m})} (0.13 \text{ T/s}) \\ &= 3.8 \times 10^{-3} \text{ V/m} = 3.8 \text{ mV/m.} \quad (\text{Answer}) \end{aligned}$$



30-7 Inductors and Inductance

an **inductor** (symbol ⌚)

can be used to produce a desired magnetic field. We shall consider a long solenoid

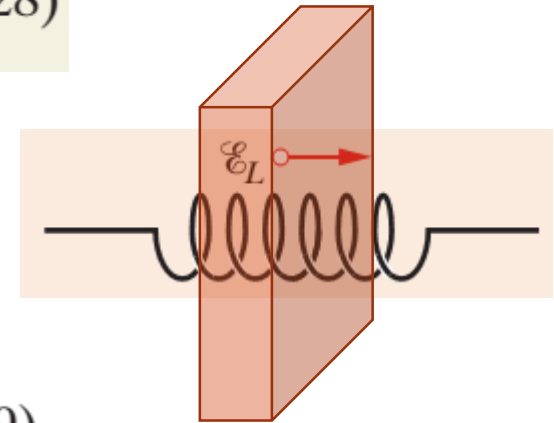
If we establish a current i in the windings (turns) of the solenoid we are taking as our inductor, the current produces a magnetic flux Φ_B through the central region of the inductor. The **inductance** of the inductor is then

$$L = \frac{N\Phi_B}{i} \quad (\text{inductance defined}), \quad (30-28)$$

in which N is the number of turns.

product $N\Phi_B$ is called the *magnetic flux linkage*.

$$1 \text{ henry} = 1 \text{ H} = 1 \text{ T} \cdot \text{m}^2/\text{A}. \quad (30-29)$$



Inductance of a Solenoid

Consider a long solenoid of cross-sectional area A .

a length l near the middle of this solenoid. The flux linkage there is

$$N\Phi_B = (nl)(BA),$$

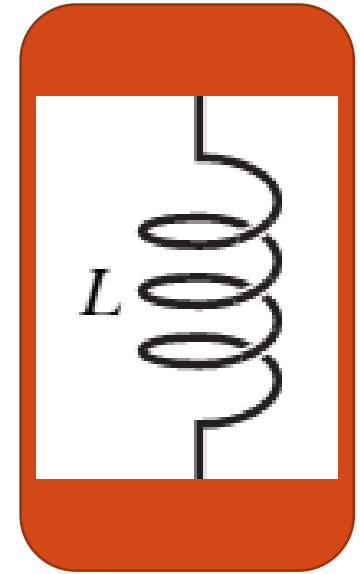
n is the number of turns per unit length of the solenoid

The magnitude B is given by Eq. 29-23,

$$B = \mu_0 in,$$

and so from Eq. 30-28,

$$\begin{aligned} L &= \frac{N\Phi_B}{i} = \frac{(nl)(BA)}{i} \\ &= \frac{(nl)(\mu_0 in)(A)}{i} = \mu_0 n^2 l A. \end{aligned}$$



$$L = \mu_0 n^2 l A.$$

Thus, the inductance per unit length near the center of a long solenoid is

$$\frac{L}{l} = \mu_0 n^2 A \quad (\text{solenoid}). \quad (30-31)$$

Inductance—like capacitance—depends only on the geometry of the device.

$$(C = \epsilon_0 A/d)$$



30-8 Self-Induction



An induced emf $\mathcal{E}_L$ appears in any coil in which the current is changing.

This process (see Fig. 30-13) is called **self-induction**, and the emf that appears is called a **self-induced emf**. It obeys Faraday's law of induction just as other induced emfs do.

$$N\Phi_B = Li.$$

(30-33)

Faraday's law tells us that

$$\mathcal{E}_L = - \frac{d(N\Phi_B)}{dt}.$$

(30-34)

Eqs. 30-33 and 30-34 we can write

$$\mathcal{E}_L = -L \frac{di}{dt} \quad (\text{self-induced emf}).$$

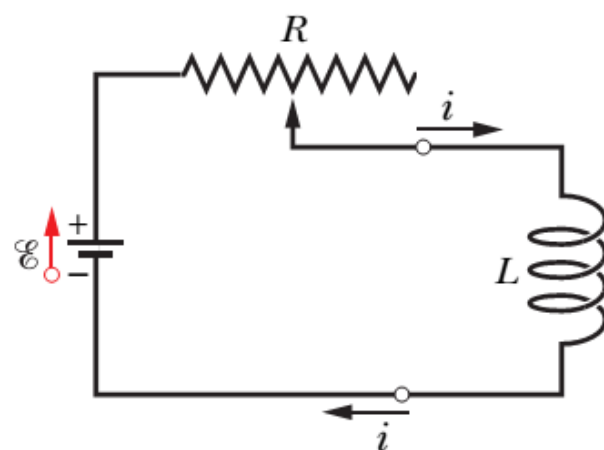
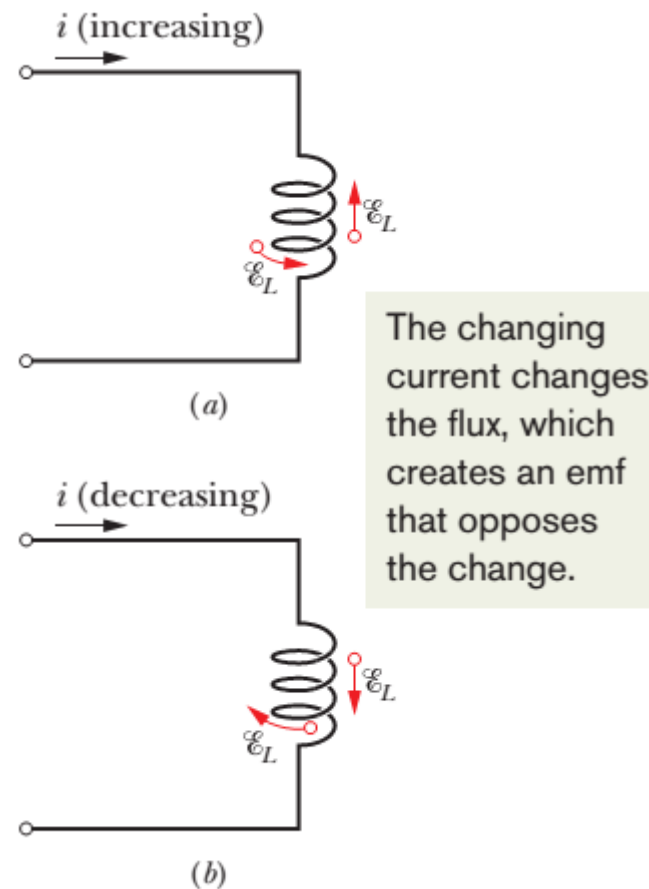


Fig. 30-13 If the current in a coil is changed by varying the contact position on a variable resistor, a self-induced emf $\mathcal{E}_L$ will appear in the coil *while the current is changing*.

You can find the *direction* of a self-induced emf from Lenz's law. The minus sign in Eq. 30-35 indicates that—as the law states—the self-induced emf $\mathcal{E}_L$ has the orientation such that it opposes the change in current i . We can drop the minus sign when we want only the magnitude of $\mathcal{E}_L$.



30-10 Energy Stored in a Magnetic Field

$$\mathcal{E} = L \frac{di}{dt} + iR, \quad (30-46)$$

$$\mathcal{E} = L \frac{di}{dt} + iR, \quad (30-46)$$

$$\mathcal{E}i = Li \frac{di}{dt} + i^2 R, \quad (30-47)$$

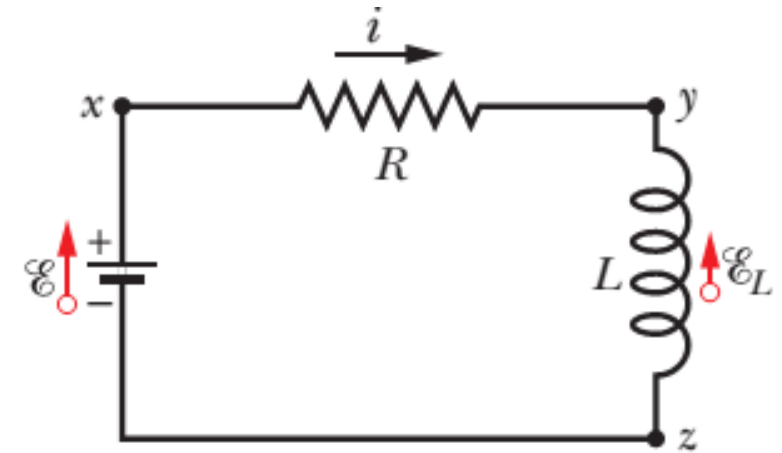


Fig. 30-16, which shows a source of emf $\mathcal{E}$ connected to a resistor R and an inductor L . Equation 30-39, restated here for convenience,



1. If a differential amount of charge dq passes through the battery of emf $\mathcal{E}$ in Fig. 30-16 in time dt , the battery does work on it in the amount $\mathcal{E} dq$. The rate at which the battery does work is $(\mathcal{E} dq)/dt$, or $\mathcal{E}i$. Thus, the left side of Eq. 30-47 represents the rate at which the emf device delivers energy to the rest of the circuit.
2. The rightmost term in Eq. 30-47 represents the rate at which energy appears as thermal energy in the resistor.
3. Energy that is delivered to the circuit but does not appear as thermal energy must, by the conservation-of-energy hypothesis, be stored in the magnetic field of the inductor. Because Eq. 30-47 represents the principle of conservation of energy for RL circuits, the middle term must represent the rate dU_B/dt at which magnetic potential energy U_B is stored in the magnetic field.

$$\frac{dU_B}{dt} = Li \frac{di}{dt}. \quad (30-48)$$

Thus We can write this as

$$dU_B = Li di.$$



$$\mathcal{E}=dU/dq \quad \& \quad \mathcal{E}=Ldi/dt$$

$$dU/dq=Ldi/dt$$

$$dU=Ldi \quad dq/dt$$

$$dU=Lidi$$

Integrating yields

$$\int_0^{U_B} dU_B = \int_0^i Li \, di$$

$$U_B = \frac{1}{2} Li^2 \quad (\text{magnetic energy}), \quad (30-49)$$

which represents the total energy stored by an inductor L carrying a current i . Note the similarity in form between this expression and the expression for the energy stored by a capacitor with capacitance C and charge q ; namely,

$$U_E = \frac{q^2}{2C}. \quad (30-50)$$

(The variable i^2 corresponds to q^2 , and the constant L corresponds to $1/C$.)



30-11 Energy Density of a Magnetic Field

we have $u_B = \frac{U_B}{Al}$ or, since $U_B = \frac{1}{2}Li^2$,

$$u_B = \frac{Li^2}{2Al} = \frac{L}{l} \frac{i^2}{2A}. \quad (30-53)$$

Here L is the inductance of length l of the solenoid.
Substituting for L/l from Eq. 30-31, we find

$$u_B = \frac{1}{2}\mu_0 n^2 i^2, \quad (30-54)$$

where n is the number of turns per unit length. From Eq. 29-23 ($B = \mu_0 in$) we can write this *energy density* as

$$u_B = \frac{B^2}{2\mu_0} \quad (\text{magnetic energy density}). \quad (30-55)$$



where n is the number of turns per unit length. From Eq. 29-23 ($B = \mu_0 in$) we can write this *energy density* as

$$u_B = \frac{B^2}{2\mu_0} \quad (\text{magnetic energy density}). \quad (30-55)$$

The equation is comparable to Eq. 25-25,

$$u_E = \frac{1}{2}\epsilon_0 E^2, \quad (30-56)$$



30-12 Mutual Induction

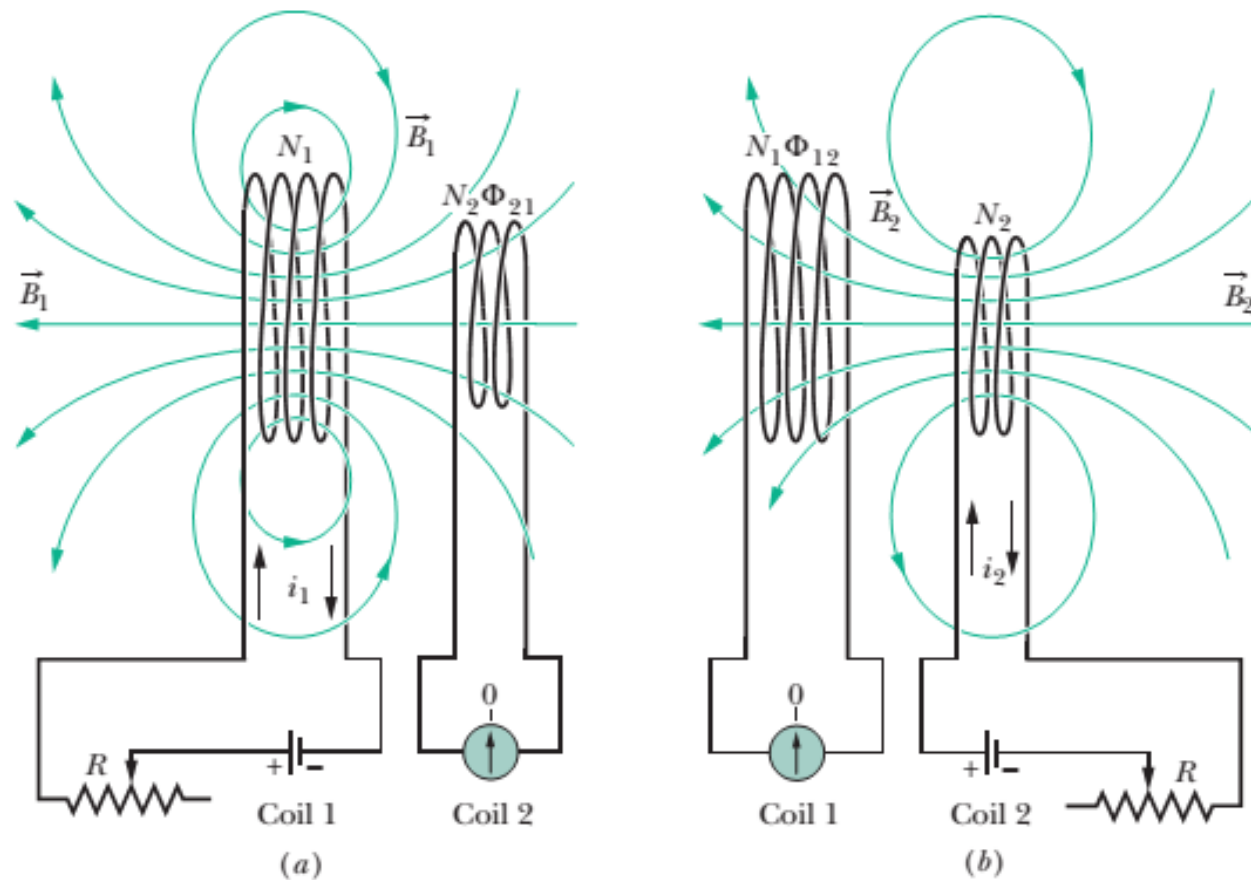


Fig. 30-19 Mutual induction. (a) The magnetic field $\vec{B}_1$ produced by current i_1 in coil 1 extends through coil 2. If i_1 is varied (by varying resistance R), an emf is induced in coil 2 and current registers on the meter connected to coil 2. (b) The roles of the coils interchanged.



which has the same form as Eq. 30-28,

$$L = N\Phi/i, \quad (30-58)$$

the definition of inductance. We can recast Eq. 30-57 as

$$M_{21}i_1 = N_2\Phi_{21}. \quad (30-59)$$

If we cause i_1 to vary with time by varying R , we have

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}. \quad (30-60)$$

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt}, \quad (30-61)$$

$$\mathcal{E}_1 = -M_{12} \frac{di_2}{dt}. \quad (30-62)$$



$$M_{21} = M_{12} = M, \quad (30-63)$$

and we can rewrite Eqs. 30-61 and 30-62 as

$$\mathcal{E}_2 = -M \frac{di_1}{dt} \quad (30-64)$$

and $\mathcal{E}_1 = -M \frac{di_2}{dt}. \quad (30-65)$



Task to do.....

Sample Problem

Mutual inductance of two parallel coils

Figure 30-20 shows two circular close-packed coils, the smaller (radius R_2 , with N_2 turns) being coaxial with the larger (radius R_1 , with N_1 turns) and in the same plane.

(a) Derive an expression for the mutual inductance M for this arrangement of these two coils, assuming that $R_1 \gg R_2$.

KEY IDEA

The mutual inductance M for these coils is the ratio of the flux linkage ($N\Phi$) through one coil to the current i in the other coil, which produces that flux linkage. Thus, we need to assume that currents exist in the coils; then we need to calculate the flux linkage in one of the coils.

Calculations: The magnetic field through the larger coil due to the smaller coil is nonuniform in both magnitude and direction; so the flux through the larger coil due to the smaller coil is nonuniform and difficult to calculate. However, the smaller coil is small enough for us to assume

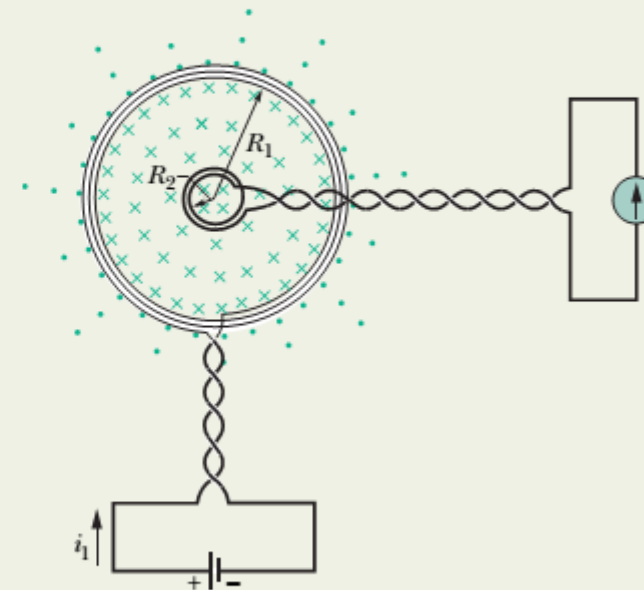


Fig. 30-20 A small coil is located at the center of a large coil. The mutual inductance of the coils can be determined by sending current i_1 through the large coil.



